Dataiku

Hands-on

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Dataiku » Learn

Learn Dataiku DSS

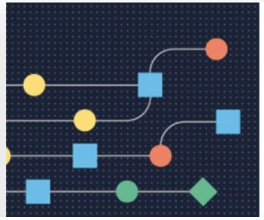
Hi, how can we help?

Start Learning Here

Course Catalog

Get Started with the Dataiku DSS Overview course and quickly move on to build more skills.

START LEARNING



On-Demand Demo

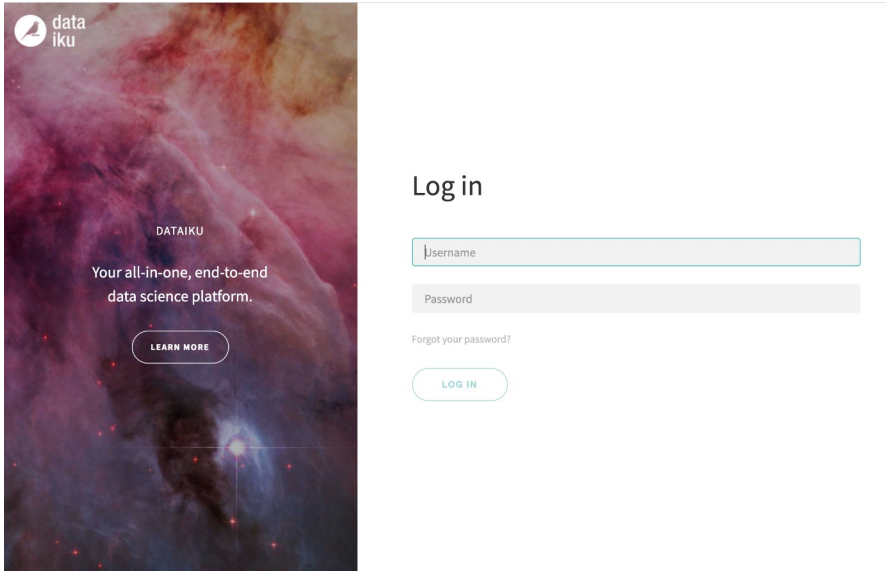
Choose your own demo with an overview based on a real-life project in Dataiku.

CHOOSE YOUR OWN DEMO

Access

http://dataiku.pjwstk.edu.pl:11000

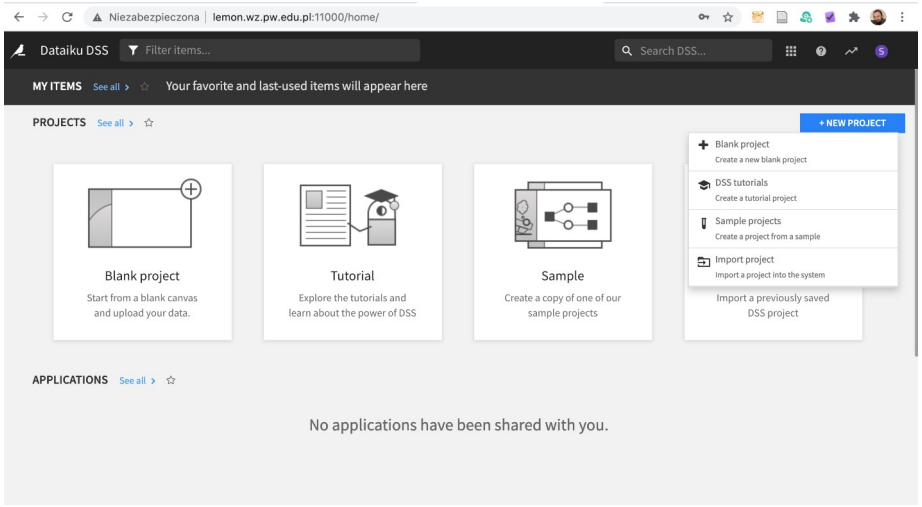
Login, pass: Lab

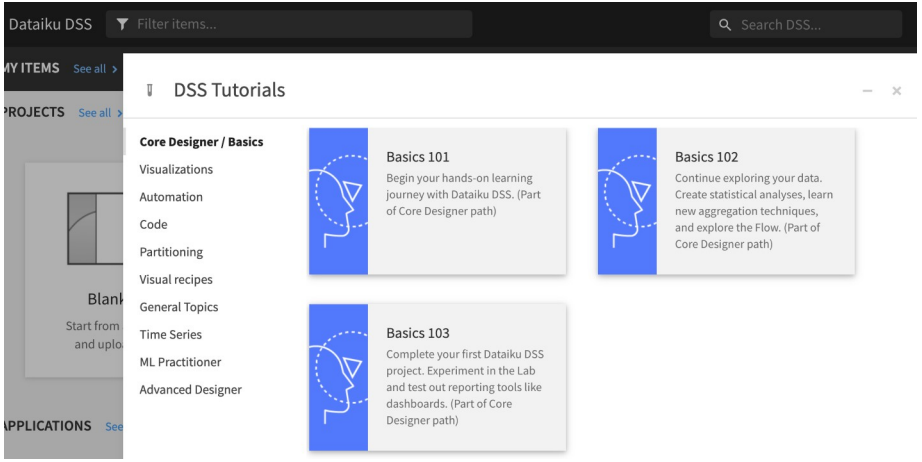


After login

New Project (top right)

- **DSS Tutorials**



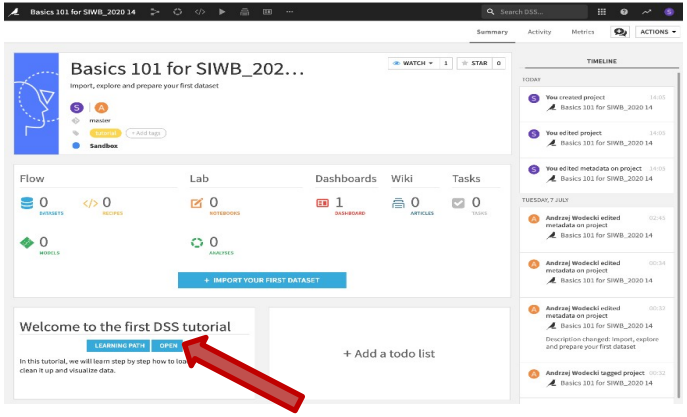


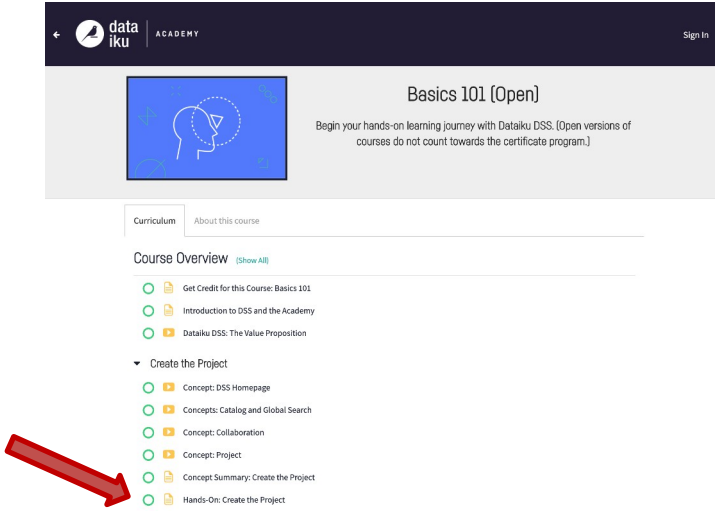
After the tutorial is created…

Welcome to the first DSS tutorial

- **OPEN**

- **Hands-On: …**





Key concepts

storing data

SCHEMA

Storage Type

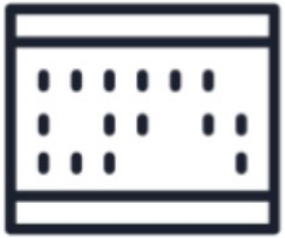
Meaning

Files | Format / Preview | **Schema** | Partitioning | Advanced

	Columns (0 / 7)	All types	All meanings	Description
1	order_date	string	Auto-detect	
2	pages_visited	string	Auto-detect	
3	order_id	string	Auto-detect	
4	customer_id	string	Auto-detect	
5	tshirt_category	string	Auto-detect	
6	tshirt_price	string	Auto-detect	
7	tshirt_quantity	string	Auto-detect	

STORAGE TYPE VS. MEANING

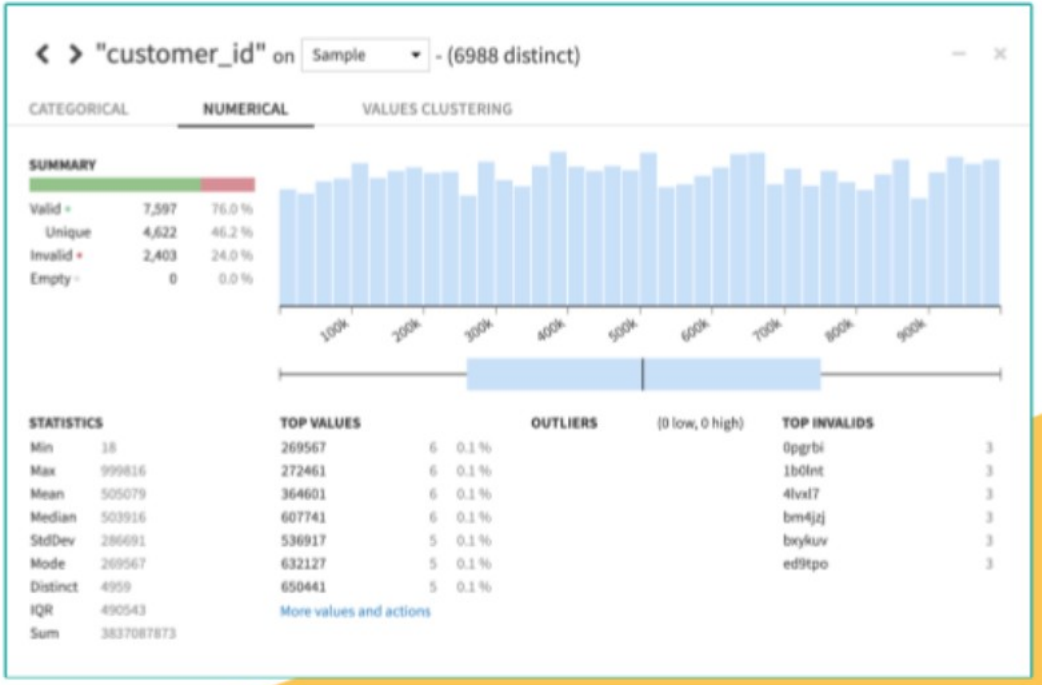
Storage Type



customer_id
string
integer
038040
801797
vftleu
914324
88ua9r
06311

Meaning

ANALYZE WINDOW



101.1: Interactive Stats

Datasets

Search DSS...

winequality

Summary | Explore | Charts | Statistics | Status | History | Settings

PARENT RECIPE | ACTIONS

Worksheet | Sampling and filtering | Confidence level: 95%

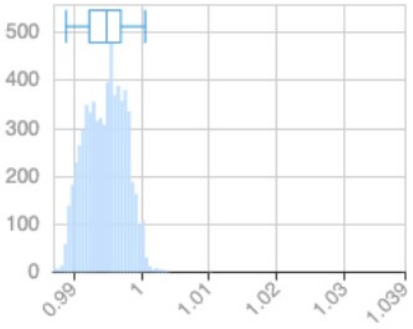
+ NEW CARD

Univariate analysis on 3 variables

No split

density

Histogram

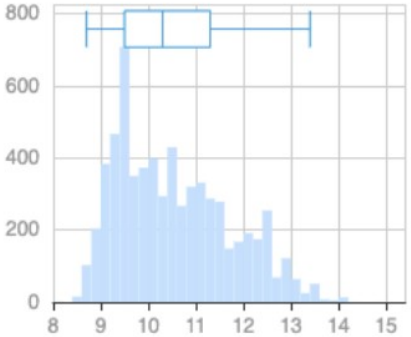


Summary stats

N values	6497
N distinct	998
N finite	6497
Mean	0.9946966338

alcohol

Histogram

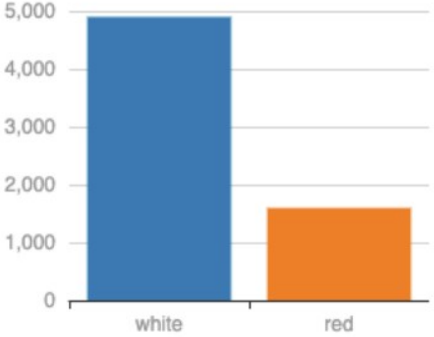


Summary stats

N values	6497
N distinct	111
N finite	6497
Mean	10.491800831

A type

Histogram

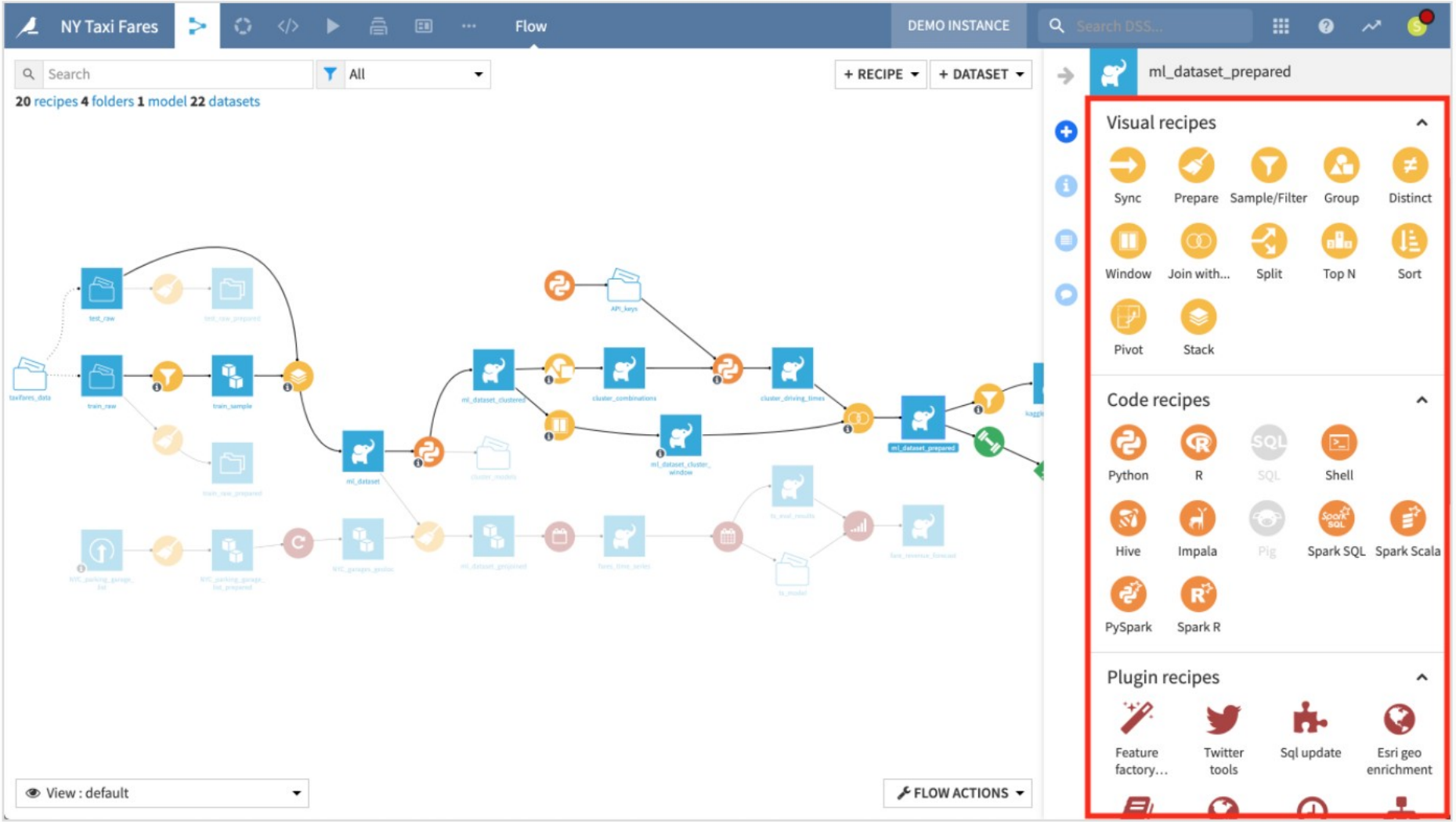


Summary stats

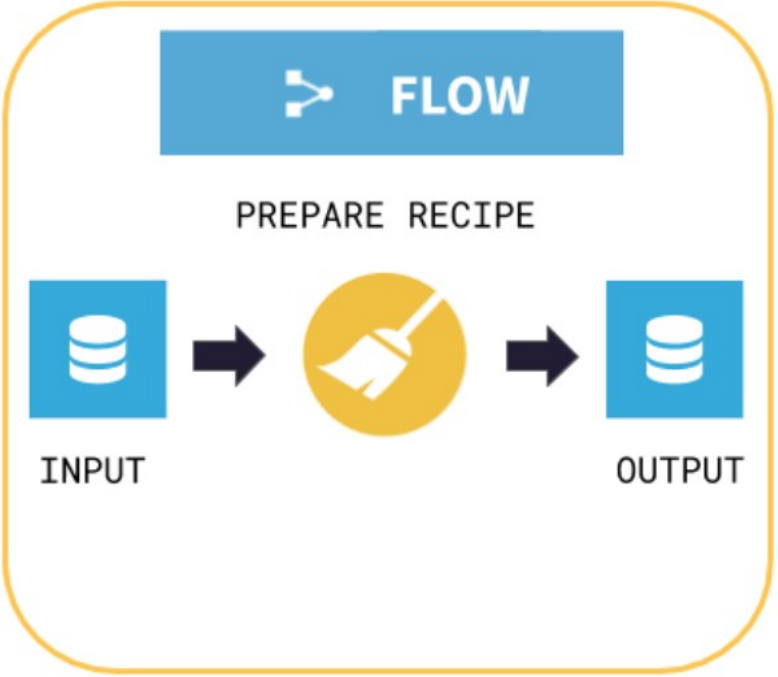
N values	6497
N distinct	2
Mode	white
N empty	0

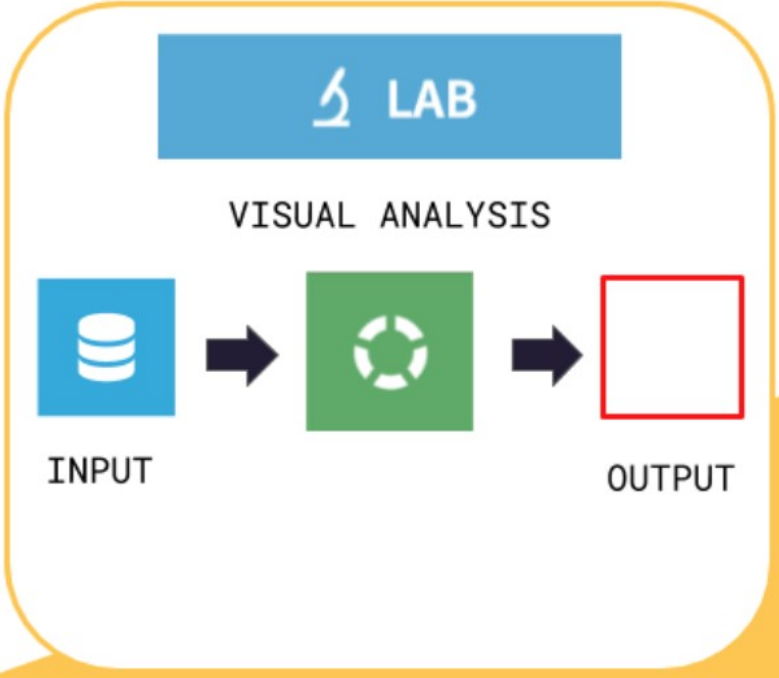
RECIPIES, FLOW

RECIPES IN DSS


NY Taxi Fares
Flow
DEMO INSTANCE
Search DSS...
Search
All
+ RECIPE
+ DATASET
20 recipes 4 folders 1 model 22 datasets
ml_dataset_prepared
Visual recipes
Sync
Prepare
Sample/Filter
Group
Distinct
Window
Join with...
Split
Top N
Sort
Pivot
Stack
Code recipes
Python
R
SQL
Shell
Hive
Impala
Pig
Spark SQL
Spark Scala
PySpark
Spark R
Plugin recipes
Feature factory...
Twitter tools
Sql update
Esri geo enrichment
View : default
FLOW ACTIONS

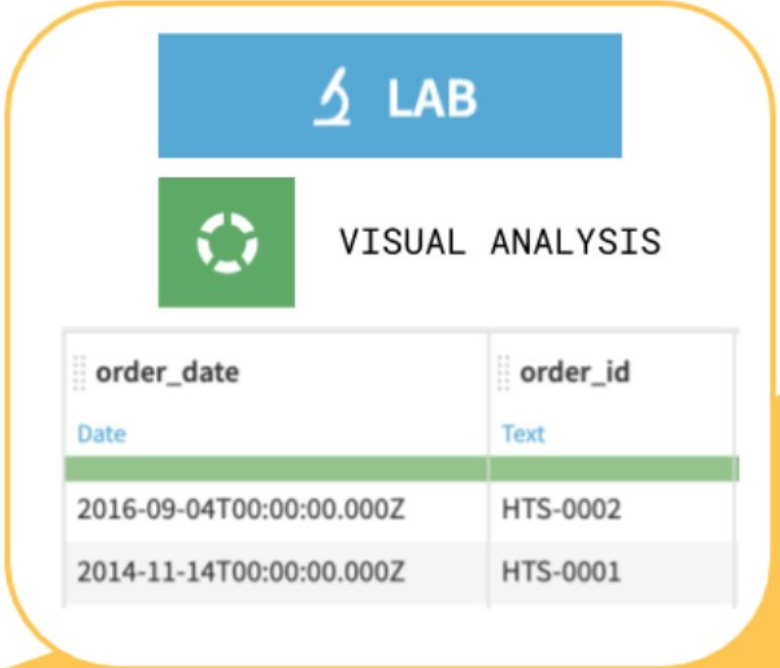
LAB VS. FLOW


FLOW
PREPARE RECIPE
INPUT
OUTPUT


LAB
VISUAL ANALYSIS
INPUT
OUTPUT

LAB VS. FLOW

FLOW
PREPARE RECIPE
order_date
string
Date
order_id
string
Text
2016-09-04T00:00:00.000Z
HTS-0002
2014-11-14T00:00:00.000Z
HTS-0001


LAB
VISUAL ANALYSIS
order_date
Date
order_id
Text
2016-09-04T00:00:00.000Z
HTS-0002
2014-11-14T00:00:00.000Z
HTS-0001

DATA PROCESSING

dates

PARSING DATES

FORMATS
“1/5/19”
“2019-01-05”
“1 May, 2019”

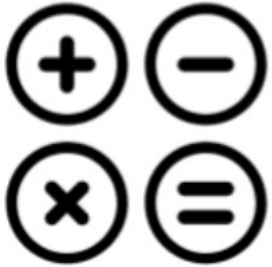
PARSED DATE
2019-05-01T00:00:00.000Z

DATA PROCESSING

group by

GROUPING COMPONENTS

1. GROUP KEY



2. AGGREGATIONS

AFTER GROUPING

DATE	CUSTOMER	AMT
2020-07-12	Carol	2
2020-07-10	Carol	3
2020-01-25	Alice	1
2020-01-05	Bob	4
2020-07-21	Alice	6
2020-10-01	Carol	5

CUSTOMER	MAX_AMT	SUM_AMT
Carol	5	10
Alice	6	7
Bob	4	4

Group Recipe Demo | Recipes | Dataiku Academy | Search DSS...

compute_orders_prepared_by_tshirt_category

Summary | Settings | Input / Output | Advanced | History | SAVED! | ACTIONS

Pre-filter

Computed columns

Group

Custom aggregations

Post-filter

Output

RUN

In-database (SQL)

Group Keys

Create a group for each unique combination of these variables

tshirt_category string

Select key to add | ADD | or create a new computed column

Compute count for each group

Per field aggregations

Select Aggregations to be computed for each field

0 / 7 | Filter column... | Hide unused variables

order_date	date	Distinct	Min	Max	~~Avg~~	~~Sum~~	Concat	~~Std. dev.~~	Count	First	Last
pages_visited	bigint	Distinct	Min	Max	Avg	Sum	Concat	Std. dev.	Count	First	Last
order_id	string	Distinct	Min	Max	~~Avg~~	~~Sum~~	Concat	~~Std. dev.~~	Count	First	Last
customer_id	string	Distinct	Min	Max	~~Avg~~	~~Sum~~	Concat	~~Std. dev.~~	Count	First	Last
tshirt_price	double	Distinct	Min	Max	Avg	Sum	Concat	Std. dev.	Count	First	Last
tshirt_quantity	bigint	Distinct	Min	Max	Avg	Sum	Concat	Std. dev.	Count	First	Last
sale_value	double	Distinct	Min	Max	Avg	Sum	Concat	Std. dev.	Count	First	Last

DATA PROCESSING

join

COMMON KEY COLUMN

ORDER#	CUSTOMER	AMOUNT
1	Carol	25.17
2	Alice	30.10
3	Daryl	18.76
4	Bob	27.31
5	Carol	22.43
6	Carol	17.32

CUSTOMER	GENDER	CITY
Carol	F	Sand Harbor
Bob	M	Boulder
Alice	F	Carson City

LEFT JOIN USE CASE

ORDER#	CUSTOMER	AMOUNT
1	Carol	25.17
2	Alice	30.10
3	Daryl	18.76
4	Bob	27.31
5	Carol	22.43
6	Carol	17.32

CUSTOMER	GENDER
Carol	F
Bob	M
Alice	F

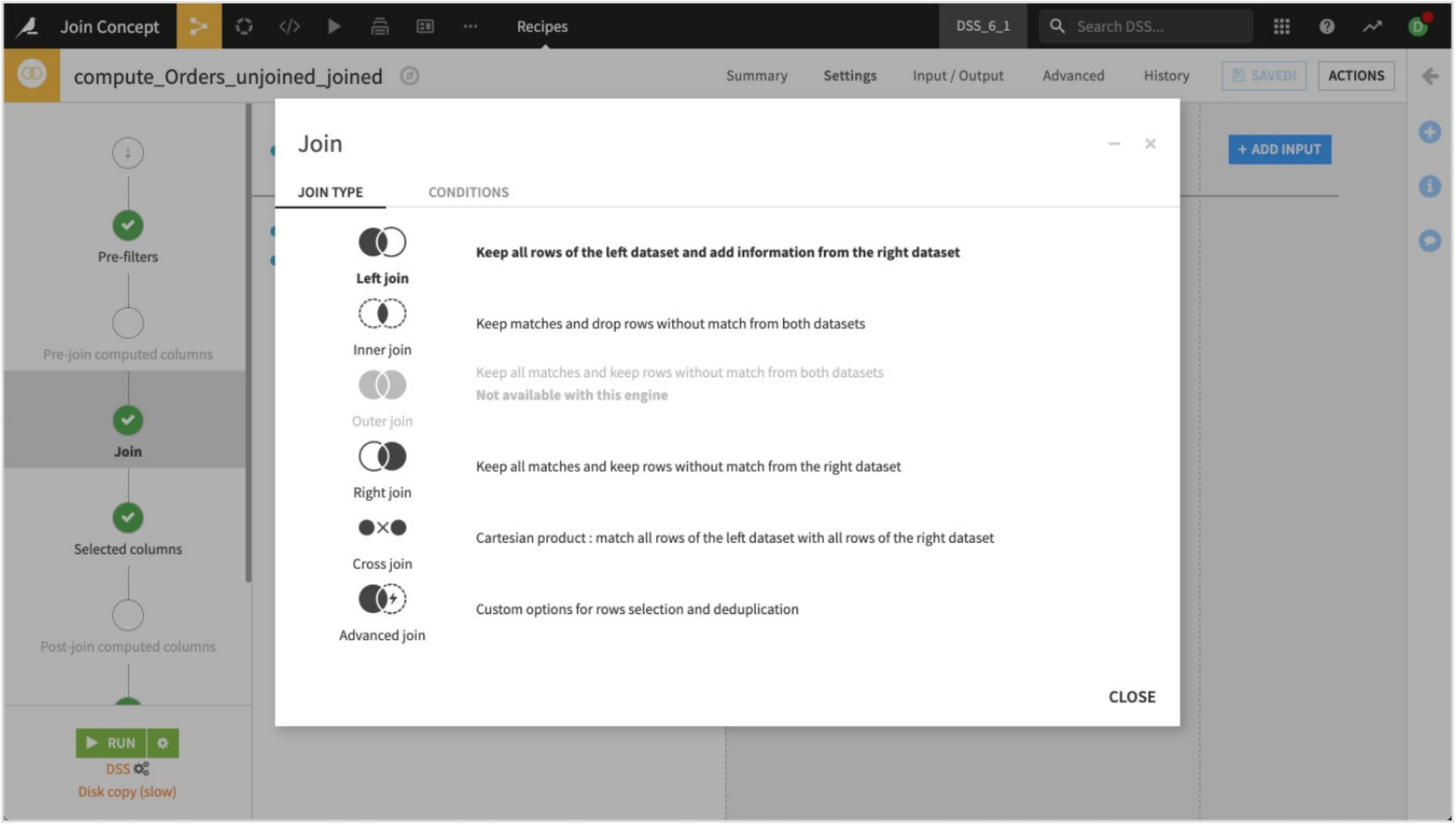
ORDER#	CUSTOMER	GENDER	AMOUNT
1	Carol	F	25.17
5	Carol	F	22.43
6	Carol	F	17.32

LEFT JOIN CHARACTERISTICS

ORDER#	CUSTOMER	AMOUNT
1	Carol	25.17
2	Alice	30.10
3	Daryl	18.76
4	Bob	27.31
5	Carol	22.43
6	Carol	17.32

CUSTOMER	GENDER
Carol	F
Bob	M
Alice	F
?	?

ORDER#	CUSTOMER	GENDER	AMOUNT
1	Carol	F	25.17
2	Alice	F	30.10
3	Daryl		18.76
4	Bob	M	27.31
5	Carol	F	22.43
6	Carol	F	17.32


Join Concept
Recipes
DSS_6_1
Search DSS...
compute_Orders_unjoined_joined
Summary
Settings
Input / Output
Advanced
History
SAVED!
ACTIONS
Pre-filters
Pre-join computed columns
Join
Selected columns
Post-join computed columns
RUN
DSS
Disk copy (slow)
Join
JOIN TYPE
CONDITIONS
+ ADD INPUT
Left join
Keep all rows of the left dataset and add information from the right dataset
Inner join
Keep matches and drop rows without match from both datasets
Outer join
Keep all matches and keep rows without match from both datasets
Not available with this engine
Right join
Keep all matches and keep rows without match from the right dataset
Cross join
Cartesian product : match all rows of the left dataset with all rows of the right dataset
Advanced join
Custom options for rows selection and deduplication
CLOSE

DATA PROCESSING

pivot

PIVOT RECIPE

NAME	ITEM	QTY	YEAR
Alice	Shoes	2	2020
Bob	Shoes	1	2017
Bob	Hat	3	2012
Alice	Hat	5	2014
Alice	Hat	2	2019
Bob	Shoes	4	2016

NAME	SHOES QTY SUM	HAT QTY SUM	SHOES YEAR MAX	HAT YEAR MAX
Alice	2	7	2020	2019
Bob	5	3	2017	2012

Concept Video: Pivot Recipe | Recipes | Dataiku Academy | Search DSS...

compute_sales_data_prepared_by_item

Summary | Settings | Input / Output | Advanced | History | SAVED! | ACTIONS

Pre-filter

Computed columns

Pivot

Other columns

Output

RUN

DSS

Examples

Simple count

Pivot table

Pivot values

Frequency table

Various statistics

Settings

Column: **year**

Row: —

Σ Content: **count of records**

id	country	year	qty
1	US	2016	7
2	US	2017	12
3	US	2017	23
4	FR	2017	8

Input

2016	2017
1	3

Count of records

1 Create columns with

item

Add new column

Pivoted values most frequent 20 values

2 Row identifiers

explicit list

name

Add new column

3 Σ Populate content with

Count of records

sum(quantity)

max(date_year)

Add new

CUSTOM AGGREGATES...

FEATURE ENGINEERING

selection, reduction, generation

FEATURE PROCESSING

NUMERICAL
FEATURES

Missing Values

NAME	ITEM	QTY
Alice	Shoes	1
~~Bob~~	~~Shoes~~	
Bob	Hat	3
Alice	Hat	4
~~Alice~~	~~Hat~~	
Bob	Shoes	4

FEATURE PROCESSING

A

CATEGORICAL
FEATURES

Categorical Encoding

NAME	ITEM	QTY
Alice	Shoes	1
Bob	Hat	3
Bob	Hat	3
Alice	Hat	4
Alice	Hat	3
Bob	Shoes	4

FEATURE PROCESSING

A

CATEGORICAL
FEATURES

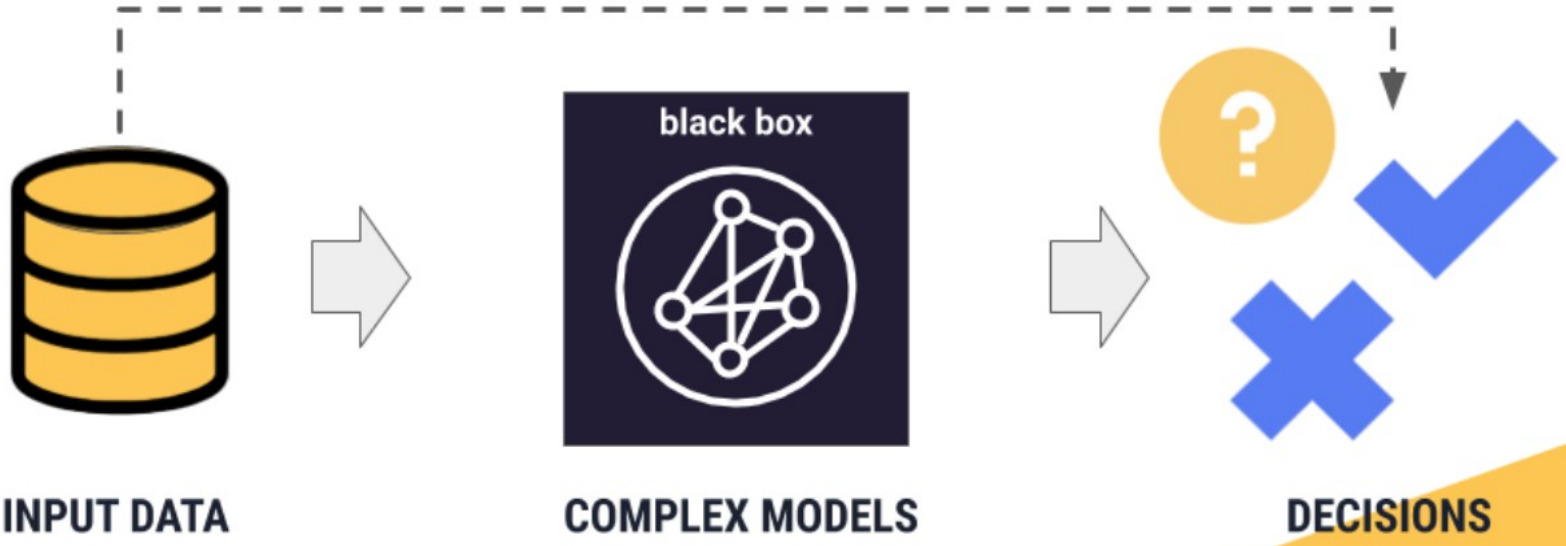
Categorical Encoding

NAME	HAT_DUMMY	SHOES_DUMMY
Alice	0	1
Bob	1	0
Bob	1	0
Alice	1	0
Alice	1	0
Bob	0	1

MODEL EXPLAINABILITY

WHITE BOX MODELS

BLACK BOX MODELS



WHITE VS. BLACK BOX MODELS

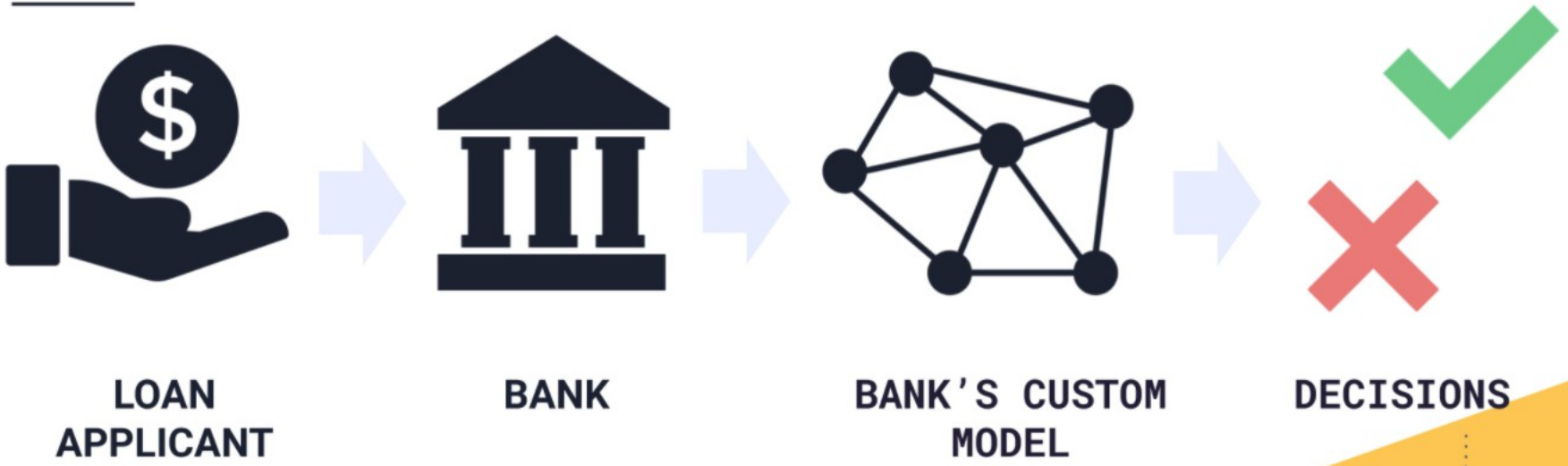
ACCURACY

EXPLAINABILITY

PARTIAL DEPENDENCE PLOT

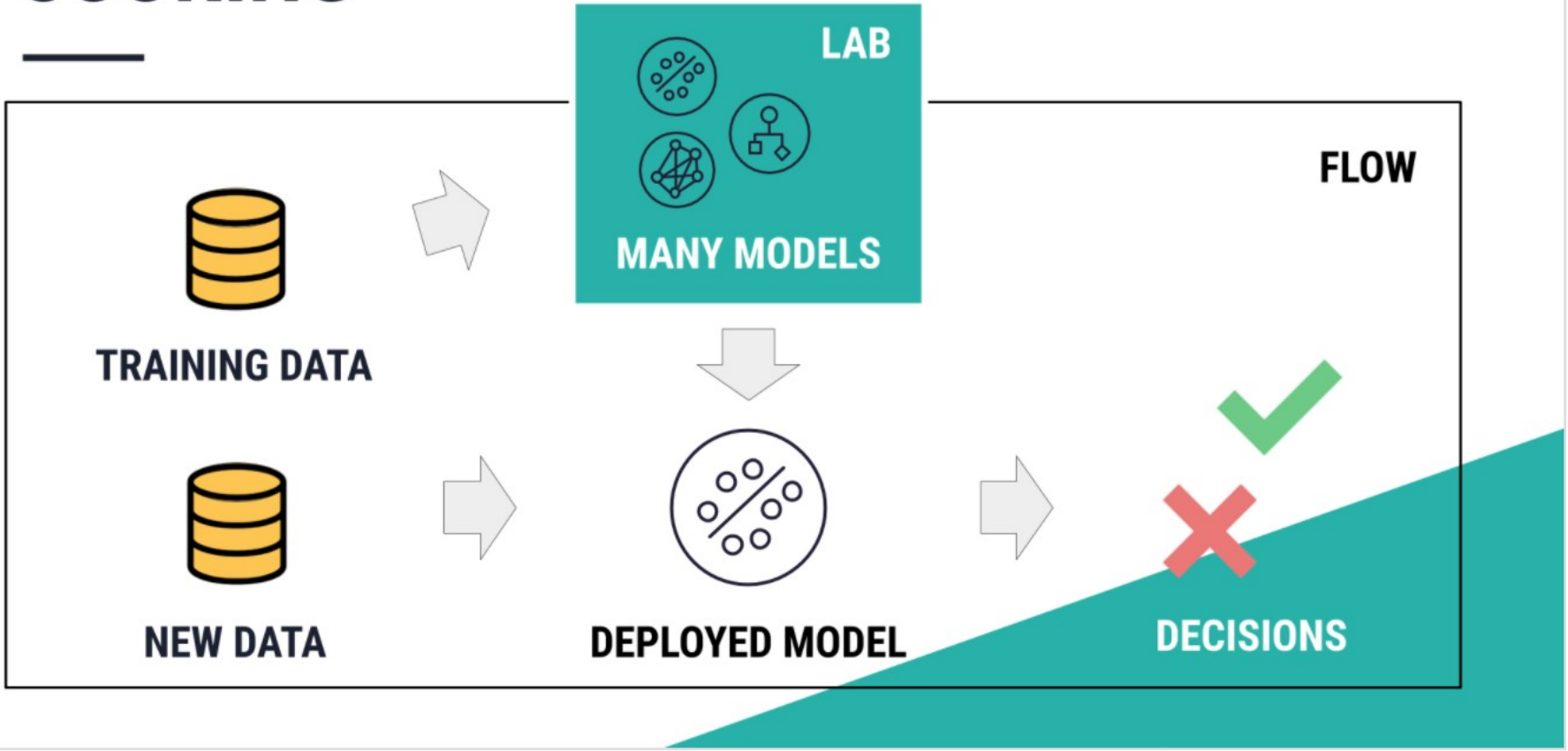
GROUP-LEVEL PERFORMANCE

INDIVIDUAL EXPLANATIONS



MODEL SCORING

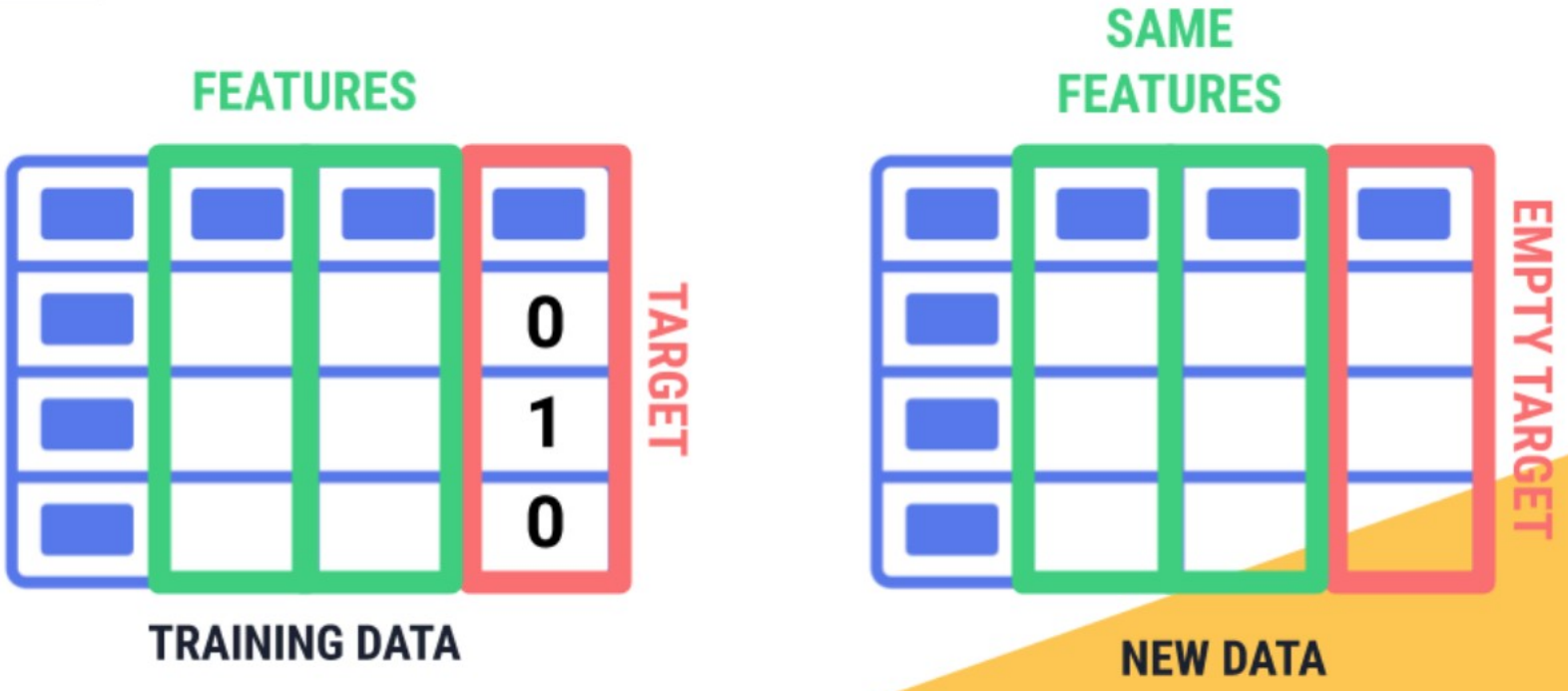
SCORING


LAB
MANY MODELS
FLOW
TRAINING DATA
NEW DATA
DEPLOYED MODEL
DECISIONS

SCORE RECIPE

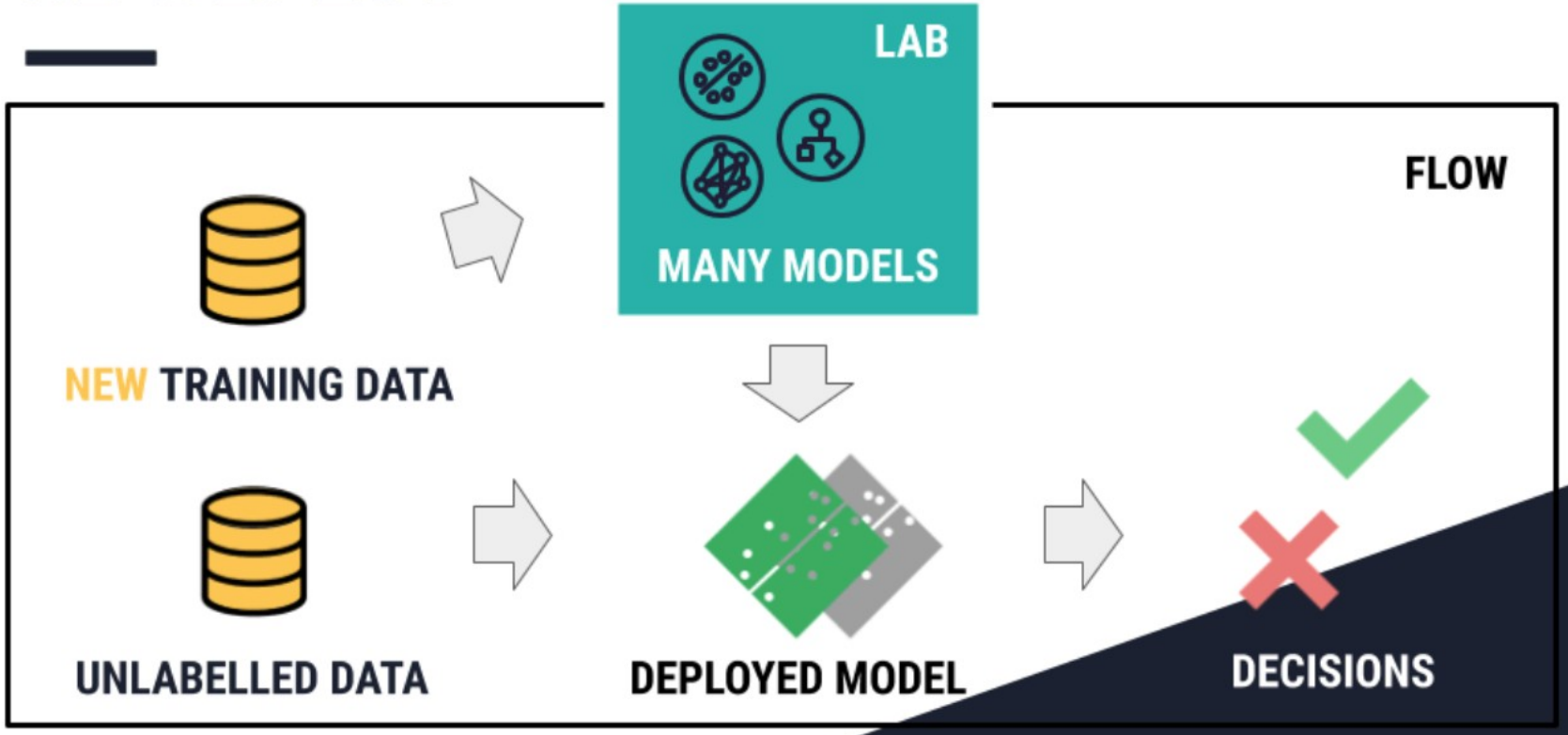
MODEL LIFECYCLE

PREPARE UNLABELLED DATA



WHEN TO RETRAIN THE MODEL?

RE-DEPLOY


LAB
MANY MODELS
FLOW
NEW TRAINING DATA
UNLABELLED DATA
DEPLOYED MODEL
DECISIONS

NEW LABELLED RECORDS